

PROGRAMMING LANGUAGES

Programming Languages

What is a ProgrammingLanguage?

- A programming language is a formal computer language designed to communicate instruction to a machine, particularly a computer.
- Programming language can be used to create programs to control the behavior of a machine or to express algorithms.
- Programming Languages can be categorized as shown in the figure:

Classification of Programming Languages

- Basically Programming languages are divided into 2 broad categories: Low level and High Level languages

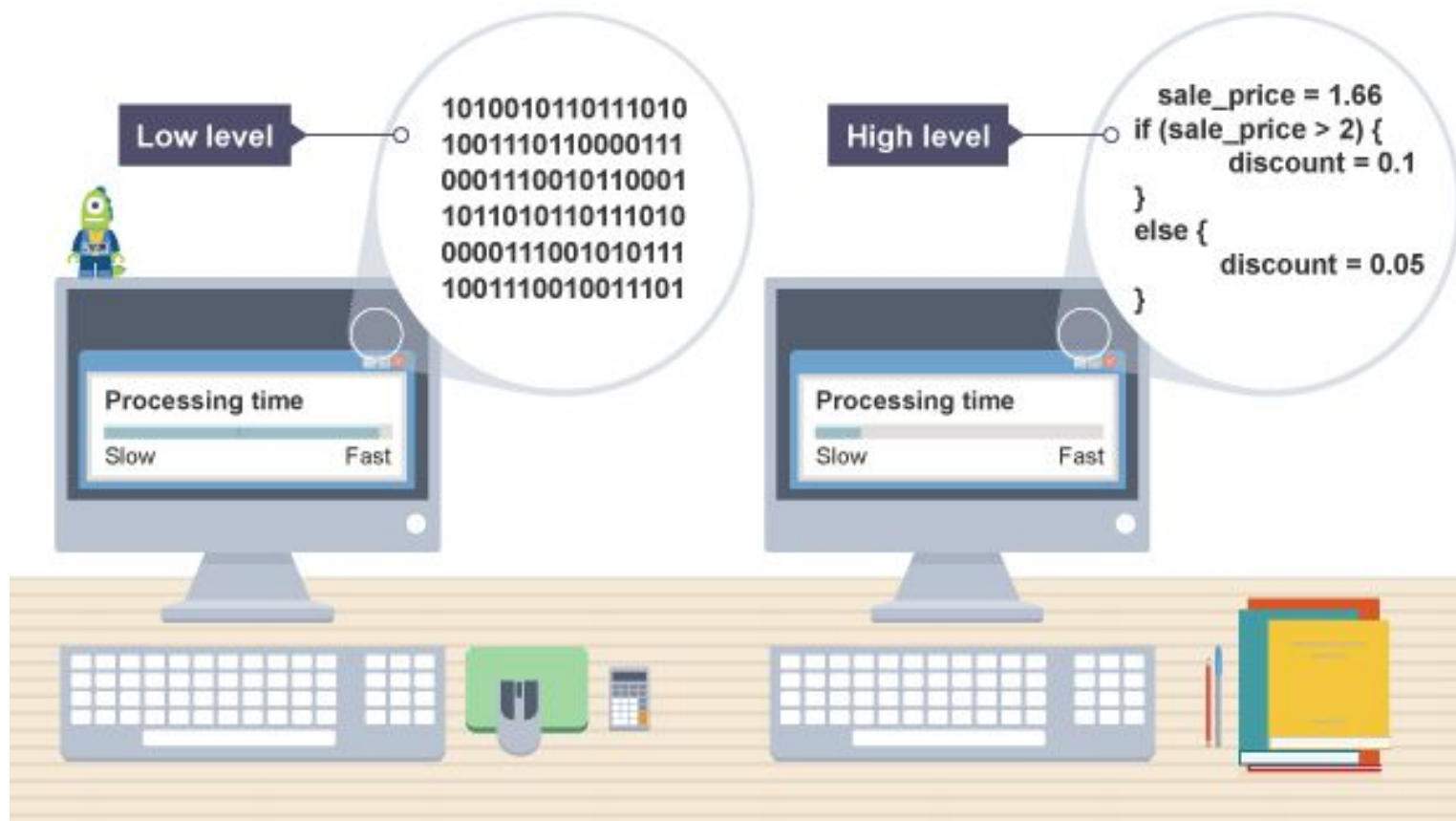
Low Level Languages

- A low-level language is a type of programming language that contains basic instructions recognized by a computer.
- Unlike high-level languages used by software developers, low-level code is often not human-readable.
- Two common types of low-level programming languages are assembly language and machine language.

High Level Languages

- Software programs and scripts are written in high-level languages, like C, C++, C#, PHP, Java, Python.
- A software developer can create and edit source code in a high-level language using a programming IDE or even a basic text editor.
- The code is not recognized directly by the CPU. Instead, it must be compiled into a low-level language.

Low Level vs High Level Programming

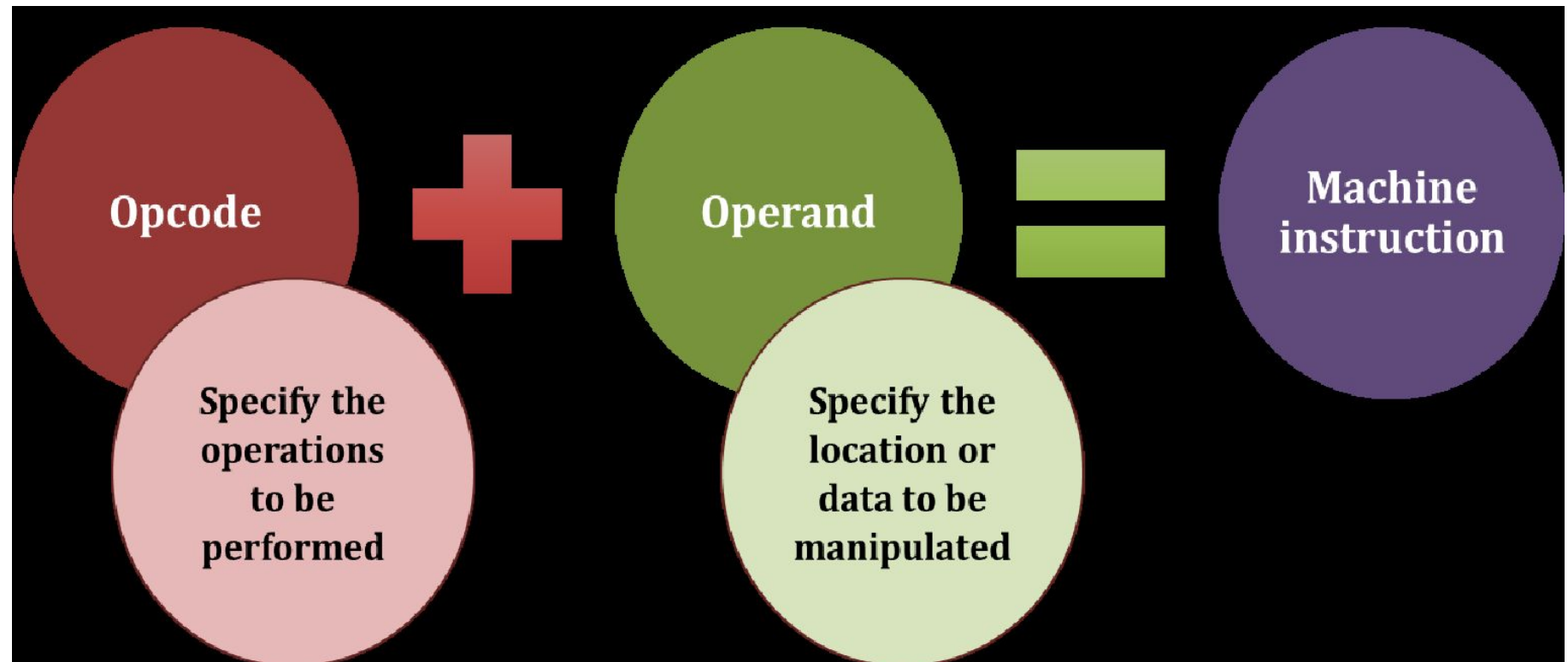


Classification of Programming Languages

First Generation or Machine Language

- ❑ Machine language, or machine code, is the lowest level of computer languages.
- ❑ It contains binary code, often generated by compiling high-level source code for a specific processor.
- ❑ Developers never need to edit or even look at machine code. Only programmers who build software compilers and operating systems need to view machine language.

Machine Language Instruction Format



Classification of Programming Languages

Second Generation or Assembly Language

- Assembly language is one step closer to a high-level language than machine language.
- Small English words like MOV (move), ADD (add), and SUB (subtract) called mnemonics are used instead of opcodes.
- These commands perform basic operations, such as moving values into memory registers and performing calculations.
- Assembly language can be converted to the machine language using an assembler.
- Assembly language is the foundation for third generation languages.
- Eg: MOV A,B
 ADD X,Y

Classification of Programming Languages

Third Generation or Procedural Language

- Refined version of Assembly language.
- Easy to write, read and understand the program.
- A programming task is broken into a collection of variables, data structures and functions known as procedures.
- Each procedure consists of a series of steps for computing.
- Programs written in high level language, i.e., English statements, cannot be executed directly, hence translators such as assemblers and compilers are used to convert them into machine language.
- Eg:- C, C++, Java, BASIC.

Classification of Programming Languages

Fourth Generation Language (4GL)

- ❑ 4GLs are focused on what is to be accomplished rather than how it is to be accomplished.
- ❑ Includes interactive coding in the form of an onscreen menu selection to formulate an enquiry.
- ❑ Easier to use and hence programmer friendly.
- ❑ Uses graphical tools and provides visual environment to work.
- ❑ To facilitate programming speed and increased productivity, high memory is needed.
- ❑ Eg:- Visual Basics (VB), SQL, Oracle, Foxpro, MS-Excel etc.

The following example shows a query in a common query language, SQL.

```
SELECT address FROM EMP WHERE empname = 'PRADIP DEY'
```

Classification of Programming Languages

Fifth Generation or Visual or Graphic Language

- Here a programmer prefers to solve problems based on the constraints provided with the program rather than using an algorithm to develop a problem.
- Application of fifth generation algorithms are mainly found in Artificial Intelligence. Eg: Mercury.

THANK YOU